Assignment 2:

Members: ulja2020 (Uliana Kulebyakina)

Group part: 73.25/80, bonus +8
ulja2020 (Uliana): 73.25 + 14.75 = 88, bonus 8 + 1 97, A

Task 1: (20/20 pts)

ok

Suggestions: `task1_generate.py` gives your files back digit for digit (900 and 300 rows, `Enhanced protection` with 270 against 3, same signs in 20.7 % and 78.7 %). The scenario and the units are clear in the README.

Task 2a: (6/10 pts)

Own function (0/4 pts): `posterior_update` in `Beta-Binomial model (task 2a).py` has no docstring, only a comment on the `def` line.
Next: a docstring that says what goes in and what comes back.

Mean and mode (3/3 pts): ok
Interval (2/2 pts): ok
Run reported (1/1 pt): ok

Suggestions: the mode line prints the formula without the case where alpha or beta is 1 or below, the README states it. The mean comes from the scipy object, the formula `alpha / (alpha + beta)` is one line.

Task 2b: (group part 8/8 pts)

ok

Suggestions: the Seaborn loader documentation is cited, and the origin of the passenger list itself is not.

Task 3a: (group part 12.25/13 pts)

Naive Bayes (4/4 pts): ok
Calculation (6/6 pts): ok
Match (2.25/3 pts): the code values 0.999515 and 0.000485 are in the README, the handwritten pages end on 0.99951471 and 0.00048529, and no line puts the two side by side. The README says that the pair “is not present in the output CSV”, which sends the reader looking for it.
Next: print the hand number next to the code number in `task3a_hand_calculation`.

Suggestions: both handwritten calculations are readable and give the same factors (0.0101339163, 0.0000049203), as the same training rows should.

Task 3b: (group part 9/9 pts)

ok

Task 3c: bonus (+8)

ok

Suggestions: 0.963 for `f1 + f2 + cat` against 0.961 for all features is a gap of 0.002, with a standard deviation of 0.012 and 0.009 over the folds. The two cannot be told apart.

Task 4a: (group part 18/20 pts)

Both models (0/2 pts): `run_experiment` and `main` in `task4_compare_models.py`, and `main` in `task3_naive_bayes.py`, have no docstring.
Next: one line each. pylint gives 9.09 on `src/`.

Suggestions: Table 8 holds every split, model and seed with the share of y=1 and the error, and split A is marked N/A, not 0. That is what the item asks.

Individual: ulja2020, Uliana (14.75/20 pts, bonus +1)

Task 2b, intervals (1/2 pts): “narrower credible interval indicates lower posterior uncertainty” has no number. The widths are 0.268 and 0.075.
Task 3, method choice (1.75/2 pts): Gaussian, why, and what it assumes are there. Independence between the features is not.
Task 3b, without smoothing (2/2 pts): ok
Task 3b, smoothing (2.5/4 pts): “introduces bias” has no number. Your table shows the fix, 0.003235.
Next: P(Enhanced protection given y=1) goes from 0 to 1/226 = 0.004425.
Task 4b, split A (1.75/2.5 pts): the 300 and 900 and the undefined sensitivity are there. What the two models did is not: Naive Bayes undefined, the tree 100 % error.
Task 4b, seeds (2/2.5 pts): 2.13 to 1.39 and 2.26 to 1.49 are right. Why stratifying lowers the spread (the test set has 25 % y=1 in every C run, 24.58 to 28.33 % in B) is not there.
Task 4b, accuracy (2.5/2.5 pts): ok
Task 4b, better model (1.25/2.5 pts): 2.17 against 2.25 % error and 98.67 against 97.67 % sensitivity are right. No feature is named: `f2` alone gives 0.947 in your 3c table.
Task 3c, bonus (1/2 pts): “additional features don’t improve classification” is more than 0.963 against 0.961 shows, a gap inside one standard deviation.

Overall

The generator gives your files back digit for digit, the maize story holds the features together, and every table of the README came back when I ran the six scripts. The Naive Bayes is your own, with docstrings, the stress test removes the three rows and says so, and the handwritten calculations are complete and agree with the code to six digits.

The group part lost points in two places that cost one line each: `posterior_update` and the two `main` and `run_experiment` functions have no docstring. In the individual texts, put a number on the cost of smoothing and name the feature behind the result in 4b, and do not claim more than the gap in 3c supports.

Running the scripts and getting every number back, as I did, makes a delivery trustworthy.